

Software Re-Engineering (SE4001)

Final Exam

Date: June 4, 2025

Course Instructor(s)

Dr. Farooq Ahmed

Total Time: 3 hour

Total Marks: 60

Total Questions: 6

Roll No

Section

Student Signature

Do not write below this line

Attempt all questions on the answer sheet. Manage time properly.

CLO 1: Describe Software Re-Engineering principles

Question 1

[5 marks]

Describe the role of metrics in conducting Software Re-Engineering exercises. How to make best use of them in order to improve decision making?

CLO 2: Explain the activities involved in Software Re-Engineering

Question 2

[5 marks]

Restructuring the architecture is an important activity in a Software Re-Engineering effort. How can Model-Driven Development approaches facilitate this aspect? Also describe possible challenges.

CLO 3: Apply different approaches to understand code

Question 3

[10+5 marks]

(a) Write BNF rules for **switch** statement in C++ language, whose syntax can be represented as follows:

```
switch(expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    ...  
    ...  
    default:  
        // code block  
}
```

Switch statement evaluates the expression. If the outcome matches with a given case value (either a variable or a constant), corresponding code block is executed. Number of case clauses can vary. A break statement can **optionally** be used to mark the boundary of the block. Moreover, a default block can also be specified **optionally** to account for scenario when no case matches the expression.

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It may be assumed that rules for expression and code block are already available.

(b) Why grammars and parsing algorithms are important in reverse engineering context? Provide proper reasoning along with example(s).

CLO 4: Compute metrics to analyze design documents

Question 4

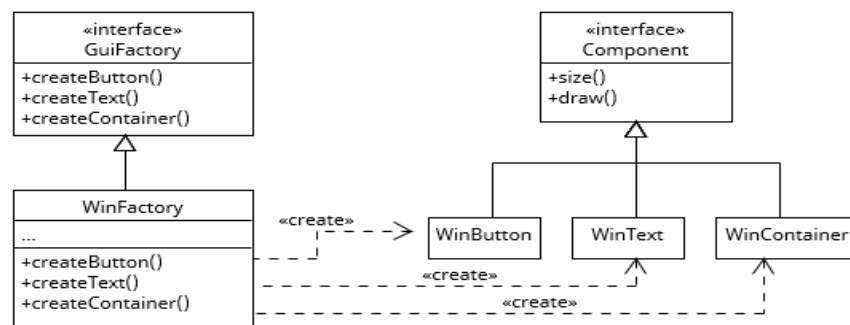
[5+5 marks]

(a) A developer computed CK metrics for two different classes with the following results:

1. class A has high DIT (> 7) but low NOC (e.g. 2)
2. class B has high NOC (> 7) but low DIT (e.g. 2)

What do you deduce from the given information about classes A and B and what potential issues can arise? Explain with proper reasoning.

(b) Consider the design below depicting application of Abstract Factory pattern:



Compute RFC for **WinFactory** class and also give your reasoning.

CLO 5: Implement refactoring strategies for effective re-engineering

Question 5

[10+5 marks]

Consider a Java program and its output below for a simple implementation of a Grid class that organizes information into a tabular format – cells organized as rows and columns. A cell in a grid is either empty or contains some value – corresponding functions are available to set and clear the cell. When a grid is printed, value of a cell is printed only if it is not empty.

<pre>public class Main { public static void main(String[] args) { Grid g = new Grid(3,5); g.setValue(0, 2, 4); g.setValue(1, 1, 7); g.setValue(1, 3, 5); g.setValue(2, 2, 1); g.setValue(2, 4, -3); g.print(); g.clearValue(1, 3); g.print(); } }</pre>	Output
	<pre> 4 7 5 1 -3</pre>
	<pre> 4 7 1 -3</pre>

```
class Grid {
    private int rows;
    private int cols;
    private int[][] cells;
    private int[][] status;

    public Grid(int r, int c){
        rows = r;
        cols = c;
        cells = new int[rows][cols];
        status = new int[rows][cols];
        for (int i=0; i < rows; i++) {
            for (int j=0; j < cols; j++) {
                status[i][j] = 0;
            }
        }
    }

    public void setValue(int x, int y, int v) {
        if (x >= 0 && x < rows && y >= 0 && y < cols){
            cells[x][y] = v;
            status[x][y] = 1;
        }
    }

    public void clearValue(int x, int y) {
        if (x >= 0 && x < rows && y >= 0 && y < cols){
            status[x][y] = 0;
        }
    }

    public void print() {
        for (int i=0; i < rows; i++){
            for (int j=0; j < cols; j++){
                if (status[i][j] == 1){
                    System.out.print( cells[i][j] + " ");
                }
                else {
                    System.out.print( " ");
                }
            }
            System.out.println();
        }
    }
}
```

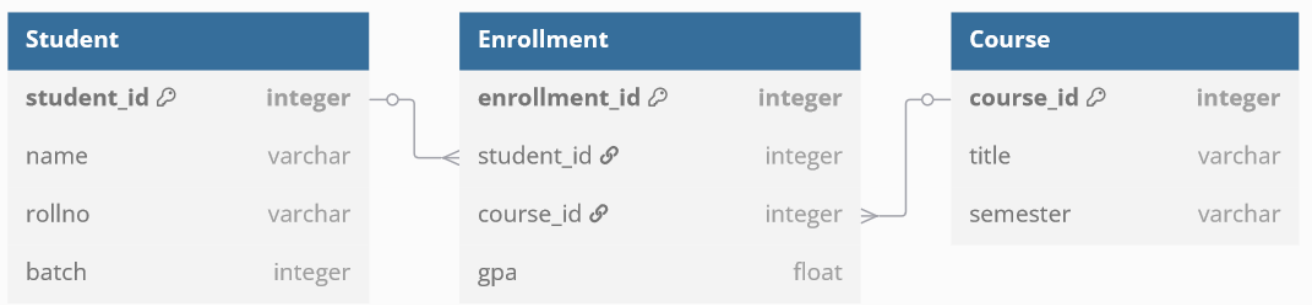
- (a) Identify the issue(s) with current **Grid** implementation and refactor (and rewrite) the code.
(b) Introduce Unit Tests for the Grid class along with refactoring.

Question 6

[5+5 marks]

- (a) Consider the following database schema for an academic registration system:

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It is noted that a significant number of queries relate to CGPA of a student. What can be done to improve the current design? Mention all the steps you will undertake to accomplish this goal.

(b) Suppose the following XML snippet captures a relationship between a child and parent class:

```
<model>
  <class name="Person" package="com.example.model" >
    <!-- other information of Person class -->
  </class>
  <class name="Student" package="com.example.model" parent="Person">
    <!-- other information of Student class -->
  </class>
</model>
```

Write XSLT script to transform XML representation into relevant code:

```
package com.example.model;

class Student extends Person {
  // other information of class goes here ...
}
```